PRANOY KOVURI

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Staff ML Engineer with 10+ years building production-scale AI systems at Apple, Google, and Amazon. Published researcher (ICSE 2026) and US patent holder. Deep expertise across **foundation model post-training** (SFT, LoRA, evaluation), **agentic and GenAI systems at million-QPS scale**, and **large-scale ML infrastructure**. Research foundations in NLP, reinforcement learning, and clinical information extraction. Technical Lead for Apple Intelligence's safety guardrail framework on 1.5B+ active devices. Track record of shipping 0-to-1 systems, including **the safety infrastructure that enabled Gemini's launch**.

CORE COMPETENCIES

LLM Post-Training & Alignment	Agentic AI & GenAI Systems	On-Device ML	Health AI · Clinical NLP	Robotics & Embedded Systems

Apple · Google · amazon · Qualcomm · PHILIPS Research

EXPERIENCE

Apple · Staff Machine Learning Engineer — Cupertino, CA · May 2025 – Present

Responsible AI / Global Safety

- **Technical Lead · Apple Intelligence Safety Guardrail Framework:** Own end-to-end on-device LoRA adapter fine-tuning, LLM-based autograding, and training data quality infrastructure spanning 17 categories across multiple languages for a 1.5B+ active device footprint.
- **Multimodal Autograder:** Drove FPR from **76% to <1%** and FNR from **26% to 9%** across 50+ controlled experiments; reframed the problem from binary classification to severity-level granularity, fundamentally changing how the team approached training label strategy.
- **Training Data Pipeline:** Auto-annotated **220K+ multimodal samples** via distributed Gemini 2.5 Pro inference across five GPU/TPU clusters; built SHA-256 cross-split deduplication that removed 5,000+ contaminated rows, directly enabling on-device adapter training.
- **Post-Training Strategy:** Benchmarked prompt-only vs LoRA fine-tuning (3B FNR: 0.29 vs 0.087), informing post-training direction; demonstrated targeted manual prompt engineering outperforms automated optimization (DSPy) across 18 head-to-head trials.
- **Platform Infrastructure:** Built unified distributed inference pipeline supporting Gemini, Apple foundation models, and Claude backends; consolidated fragmented codebase (22 configs to 1 template); authored design document for guardrail inference API serving partner teams.
- **Evaluation & Bias Audit:** Led first scaled multi-model image safety benchmark (19,834 images, 3 foundation models); built Hinglish evaluation framework for code-mixed content; executed systematic bias audit, surfacing 7 distributional biases adopted by upstream curation teams.

Amazon · Senior SDE (SDE III) — Palo Alto, CA · Jan – Apr 2025

Search Ranking

- **Deep Learning Ranking:** Led development of **ranking models for Amazon's core search engine**; engineered scalable model export and deployment pipelines for fine-tuning on distributed GPU clusters; worked on final-stage ranking determining product relevance for hundreds of millions of customer queries.

Google · Software Engineer – Machine Learning — Sunnyvale, CA · 2019 – 2024

Cloud Platform Safety Filters, Vertex AI

- **Vertex AI Safety Architecture:** Architected the safety filter serving layer for Vertex AI's generative products; designed routing and classification (SafetyCat, 350M-param multilingual model) gating all model outputs across 18 categories with severity scoring at production scale.
- **Configurable Filter Granularity:** Defined block_few/block_some/block_most modes; achieved >90% precision in default mode and a 95% reduction in over-blocking, directly enabling enterprise Cloud AI adoption.

- **Canonical Protections Integration:** Led integration of GRAIL ensemble classifiers serving 100M+ content pieces/week across 10+ product areas, unifying fragmented safety infrastructure under a single classification interface.

AI Content Safety, Bard/Gemini Initiative

- **GRAIL Classification Service:** Built and launched a self-service agentic safety platform serving 1M QPS across 10K+ classifiers for all Google generative AI products; designed and shipped in one month, adopted by 50+ clients including Gmail, Bard, DeepMind, and Cloud API.
- **Multilingual Training Pipelines:** Built a Gemini-based agentic translation pipeline processing 100K data points in 5 minutes, scaling training data to 140 languages; eliminated $15M/month in inference API costs.
- **Crisis-Response Classifier:** Designed and deployed a terrorist content safety classifier in one month during active conflict; improved precision from 1% to 82% using distilled AutoMUM models across 75 languages.
- **Bard / Gemini Code Red:** Curated 200M labeled entities across 10+ products, built LLM-based moderation workflows, improved reviewer efficiency 6x; created model factory platform for rapid LLM-powered classifier development.

Text Content Safety, Canonical Protections

- **Text Deobfuscation System (US Patent):** Invented and deployed at 1B+ QPS across YouTube, Gmail, Drive, and Messaging; 400% improvement in mapping coverage with sub-millisecond latency.
- **Spam Detection:** Advanced from 25P@90R to 98P@90R through interpretability-guided data cleaning; launched real-time abuse detection across Hangouts, Messages, Voice, and Chat.
- **Online Evaluation Framework:** Built live-traffic duplication and differential metrics, compressing model rollout cycles from months to days and reducing manual review cost 10x.

Philips Research · Research Intern (Health AI) North America · Summer 2018

- **Clinical NLP:** Designed deep learning pipelines for entity and relation extraction from chest X-ray radiology reports, building structured knowledge bases for downstream ailment prediction.

Texas A&M University · Research Assistant College Station, TX · 2017 – 2019

NLP, Reinforcement Learning, Clinical Information Extraction

- **Joint Entity & Relation Extraction:** Developed neural architectures using custom Transformer pretraining; work influenced subsequent lab research on clinical report generation (EMNLP 2020 Findings).
- **Reinforcement Learning Research:** Designed and tested Q-Learning, DDQN, DDPG, and IRL algorithms for a novel research problem with custom gym environment design; presented RL research papers internally at Google.
- **Clinical Outcome Prediction:** Improved ICU readmission prediction by incorporating NLP-derived features from clinical notes; Teaching Assistant for NLP and Senior Capstone Design.

Qualcomm · Software Developer Hyderabad, India · 2014 – 2017

Wi-Fi Embedded Systems

- **Embedded C/C++ Systems:** Developed real-time features for Wi-Fi subsystems; provided onsite engineering support for OEM product launches at Qualcomm China; recognized with 5 Qualcomm Stars awards.

PUBLICATIONS & PATENTS

- S. Jiang*, **P. Kovuri***, D. Tao, Z. Tan. "CASCADE: LLM-powered JavaScript Deobfuscator at Google." **ICSE SEIP 2026** (48th IEEE/ACM International Conference on Software Engineering). *Equal contribution.
- **US Patent:** "Detecting a Homoglyph in a String of Characters" (US20240346840A1); unified detection system for adversarial text obfuscation attacks.
- **Master's Thesis:** "End-to-End Relation Extraction using Semi-Supervised Pretraining." Texas A&M University, 2019.

AWARDS & RECOGNITION

- **Google Peer Bonus (2023):** Leadership in Safety Data infrastructure for Gemini's safe launch.
- **Google Spot Bonus (2023):** Hate-speech detection model launch with 86% reduction in false positives across Google client platforms.
- **Google Spot Bonus (2021):** Lead developer on the unified Character Deobfuscation Library deployed across YouTube, Gmail, Drive, and Chat.

PROFESSIONAL SERVICE & SPEAKING

- **Conference Reviewer:** ACL 2024-2025 (7 manuscripts, top-tier NLP), IJCNN 2025-2026 (5 manuscripts).
- **Invited Speaker:** "Use of Agents in Gen AI" — qAIntum.ai Quantum Gen AI Lecture Series (2024); SJSU "Exploring Responsible Computing Careers" panel (2025).

EDUCATION

Texas A&M University · MS in Computer Science (specialized in AI), GPA 4.0   2019

NIT Warangal · BTech Electronics & Communication Engineering, GPA 3.8   2015

TECHNICAL SKILLS

Languages: Python, C/C++, Go, JavaScript/TypeScript

ML Frameworks: PyTorch, TensorFlow, JAX, Hugging Face Transformers

Specializations: Foundation model post-training (SFT, LoRA), agentic systems and multi-model orchestration, transformer architectures, prompt engineering, RAG, evaluation framework design, distributed training, on-device ML, model distillation, reinforcement learning (research)

Domains: Generative AI, NLP, content safety & alignment, multilingual ML, clinical/health AI, search & ranking, embedded systems